P∞-G: The Cubic Exclusion — Order-Three Arithmetic Against Marginal Boundary Contact of the Seam Renewal

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Position in the series. This paper consumes P∞-F §§19–24 (the center-out chain F.21, the energy ledger F.22, the transfer map and its certification F.23, the comparison/non-expansiveness lemma F.24, the conditional closure F.25–F.26, and the parity factorization F.27) and executes the lemma queued since P∞-D §7(c): *stationarity at a fixed point forces Gaussianity; the program's cubic theorems become load-bearing.* It is, deliberately, a different **type** of strike from everything in the F-series: not a reorganization of window Weil positivity (every order-2 reorganization is pre-killed by the program's own blindness theorems — CBS at order 2, the Tower theorem, P∞-B Theorem B.1), but a **consumption attempt**: spend an unconditional order-three sign — the cubic chirality ($c_2 < 0$ via $L(\frac{1}{2}, \chi_3) = 0.4808\ldots > 0$; the non-real Jacobi phases of P∞-B Theorem B.2; the cumulant transfer $\tau_3 = -1$ of N5 Theorem 6.2) — to constrain the *asymptotic geometry* of the seam renewal. The audit of what such a strike can and cannot prove is §2, placed before any mathematics that could be motivated by hope.

1. The inherited question, and the trichotomy

By Theorem F.21, RH $\iff$ $d(s) \geq 0$ for every s — the center-out innovation density of the one chain stays nonnegative, each value a finite exact computation, permanent once certified. By F.23–F.26 the chain reduces, up to the certified bulk collapse ($c_{\text{clean}} \approx 2.7$–$3.8$ per unit λ^2), to a **seam-renewal map**: the boundary cluster's conditional state, renewed once per unit λ^2 by the PNT-paced, Fejér-ramped arrival stream, with the unpredictable remainder carried by the single even seam cell (F §23.2; the parity factorization F.27 makes the even sub-chain the carrier and the odd contribution a perfect square). The measured orbit obeys the RG-time power law $R_{\text{seam}} \propto \lambda^{-p}$, $p = 3.2 \pm 0.2$ (F §23.6), whose local slope at the data centroid is the §21 constant $c = 0.342$.

The asymptotic alternatives for the seam orbit are three:

(1) Exit. Some pivot crosses zero at finite radius: RH is false (F.21). **(2) Uniform interiority.** Hypothesis (U) of F.25: the renormalized boundary states stay in a Thompson ball interior to the cone; then the classical contraction engines give the stationary profile, the law, and RH — *with room.* **(3) Marginal contact.** The orbit stays in the cone but approaches its boundary in renormalized scale — "RH true, but barely": positivity surviving with asymptotically degenerating margins. A wandering variant (boundary grazing along a subsequence without convergence) belongs to this class.

Scenario (3) is the dangerous one, for two reasons. First, it is the scenario the architecture cannot certify against by finite computation: every certified window is consistent with margins that die later. Second, the measured deceleration of the remainder law's local slope ($0.547 \to 0.171$ across the seven windows) is *exactly what an approach to marginality would look like* — it is also exactly what the lawful power law looks like, and distinguishing the two is the open step's quantitative face (F §23.6: "$p < \infty$, one exponent"). This paper attacks scenario (3) directly.

2. The logic audit: what the cubic can and cannot do

The category guard. The renewal map T is built from the window Weil kernel; its state-derivatives DT, D^2T, D^3T are rational functions of kernel entries — **order-2 arithmetic data**. No Taylor coefficient of T in the state contains a third cumulant of anything. The genuine order-three arithmetic enters through the τ**-dependence of the map**: which prime powers arrive when — the deviation of the arrival stream from its mean. Operationally (§4): the directional expansion of the seam pivot along the **Cramér fluctuation** $\Delta K = K_{\text{true}} - K_{\text{Cramér}}$, the difference between the true kernel and the kernel with its prime atoms replaced by the PNT-smooth density. By the explicit formula, ΔK *is* the zero-oscillation content of the kernel — computed entirely on the prime side, with no zero input. Any claim that "the cubic excludes X" must route through this channel or it is a category error; the first draft of this strike (the launch prompt's "boundary fixed profile forces the stream's third cumulant to vanish") was repaired into the present form by exactly this audit.

What success proves. If the windowed cubic drift along the contact asymptotics is bounded away from zero (Proposition G.2 + Theorem G.3), then *asymptotically stationary marginal boundary contact is impossible*: the trichotomy collapses to the dichotomy {exit, interior with room}. Consequences: (i) "RH true but marginally" dies — if positivity holds, it holds with room, discharging the orbit-version of the degeneracy face (the (QS_{exp})-adjacent worry that gap statistics degenerate along the chain); (ii) the law's deceleration must terminate at an interior profile — **the exponent** p **must stabilize**, a falsifiable prediction that the already-registered $\lambda = 3.5$ discriminator (F §23.6) now doubly tests; (iii) F.25's contraction applies on the interior side unconditionally of marginality worries.

What success does not prove. The exit scenario — RH false — is untouched: the RH-false world contains the same primes and the same cubic, and nothing here constrains a clean crossing. (The exclusion constrains *stationary boundary* behavior; a crossing is not stationary.) The wandering variant is constrained only modulo a convergence input, for which the measured correlation-orbit settling (F §23.6, X2→Y2) is evidence, not proof; this caveat is carried explicitly in G.3. The conserved core is conserved: what this strike buys, if it stands, is the *removal of the marginal corridor* between the two clean alternatives, plus a sharp empirical prediction.

W2-compliance. Every decision rule below fires on instrument or route failures (sign instability, decoupling, reconstruction breakdown), never on the RH-true world: the true kernel's orbit is interior at every certified window, and all registered falsifiers are statements about the *bridge object*, not about positivity.

3. Lemma G.1: one-sided dynamics at a marginal boundary fixed point [P]

Lemma G.1. *Let $P \subset \mathbb{R}^N$ be a closed convex cone, $x^**\in \partial P$, and ν a supporting functional at x^** ($\nu \geq 0$ on P, $\langle \nu, x^** \rangle = 0$). Let T be C^3 near x^** with $T(P \cap U) \subseteq P$ and $T(x^**) = x^**$. Call w admissible if $x^* * + \varepsilon w \in P$ for all small $\varepsilon > 0$.*

(i) $\langle \nu, DT(x^**)w \rangle \geq 0$ for every admissible w.

(ii) If v is admissible and ν-marginal — $\langle \nu, DT(x^**)v \rangle = 0$ — then $\langle \nu, D^2T(x^**)[v,v] \rangle \geq 0$.

(iii) If $\pm v$ are both admissible (a tangential mode) with $\langle \nu, DT(x^**)v \rangle = 0$ and $\langle \nu, D^2T(x^**)[v,v] \rangle = 0$, then

$$\langle \nu,\ D^3T(x^**)[v,v,v] \rangle \ =\ 0 :$$

odd-order pushes along a tangential marginal mode must vanish at a boundary fixed point.

Proof. (i) $0 \leq \langle \nu, T(x^* * + \varepsilon w) \rangle = \varepsilon \langle \nu, DTw \rangle + O(\varepsilon^2)$; divide by $\varepsilon \downarrow 0$. (ii) With the linear term dead, $0 \leq \langle \nu, T(x^* * + \varepsilon v) \rangle = \frac{\varepsilon^2}{2} \langle \nu, D^2T[v,v] \rangle + O(\varepsilon^3)$. (iii) With linear and quadratic terms dead, $0 \leq \frac{\varepsilon^3}{6} \langle \nu, D^3T[v,v,v] \rangle + O(\varepsilon^4)$ for $\varepsilon > 0$ gives ≥ 0; replacing $v \to -v$ (admissible) gives ≤ 0. □

G.1′ (orbit version) [P**-sketch**]**.** For a τ-dependent family T_τ with no exact fixed point — our case — the same one-sidedness applies along the orbit: if $x_\tau \to x^** \in \partial P$ boundary-marginally (renormalized

distance $\to 0$) with asymptotic approach mode v, then the **windowed drift** of the boundary distance, $\frac{1}{|W|}\sum_{\tau\in W}\langle \nu, \mathrm{push}_\tau\rangle$, is ≥ 0 and $\to 0$; expanding push_τ in the state and in the arrival fluctuation, the leading non-vanishing windowed coefficient must vanish asymptotically. Contrapositive: *a windowed push coefficient bounded away from zero along the approach data forbids the approach.* The sketch's gap, stated honestly: the identification of the approach mode and the uniformity of the window averages — these are exactly what the certificate measures and what G.2's analytic weld must supply.

Remark (the two-sidedness is unconditional — Littlewood supplies it). In G.1(iii) the hypothesis "$\pm v$ both admissible" looks like a geometric assumption; in the orbit version G.1′ its role is played by the *sign-exploration of the fluctuation channel*, and there it is a theorem: $\psi(x) - x = \Omega_\pm(x^* 1/2 \log\log\log x)$ (Littlewood 1914), unconditionally — the arrival stream's deviation from its Cramér mean changes sign infinitely often at the $\sqrt{}$-scale, which is precisely the scale the $n^{-1/2}$-normalized kernel reads. Even-order push terms, being sign-definite in the fluctuation, cannot cancel an odd-order term whose argument provably oscillates: at a stationary marginal contact the *windowed odd coefficient itself* must vanish. The exclusion's structural half therefore consumes only unconditional inputs (cone preservation + Littlewood); the entire conditional load sits in G.2's identification of that coefficient with φ_3 along the contact asymptotics.

Parity bookkeeping (symmetry-killed channels). The kernel is exactly reflection-invariant (certified at 10^{-49}, F §24), so T is reflection-equivariant. If v is a *pure odd* mode and ν even, then $D^3T[v,v,v]$ is odd and $\langle \nu, D^3T[v,v,v]\rangle = 0$ **by symmetry** — a vanishing with no arithmetic content. The surviving cubic carriers are the even modes and, decisively, the τ-fluctuation channel of §4; the bridge routes the arithmetic through the latter. (This guard matters: a naive "compute the cubic along the odd tower" would measure a symmetry zero and report decoupling.)

4. The bridge object: the Cramér expansion of the seam pivot

The Cramér base. Replace the prime block of the certified builder, $\sum_n \Lambda(n) n^{-1/2}\varphi(\log n)$ over prime powers $\log n \leq L$ (Fejér-damped as in F.1's multiplier), by its PNT-smooth partner $\int_0^L e^{u/2}\varphi(u)\,du$ — the unique density making the §18/F.19 pole-matching exact term by term (the Cramér world's explicit formula has a pole and **no zero oscillations**). Write $K_{\mathrm{true}}, K_{\mathrm{Cram}}$ for the two point-basis kernels and $\Delta K = K_{\mathrm{true}} - K_{\mathrm{Cram}}$: the **zero-oscillation content of the kernel, computed prime-side**. Two structural facts: ΔK vanishes in the Cramér world (tautologically), and the $\mathcal{E}$-ladder mechanism (E.22.1's radical membership) requires the Euler atoms — the Cramér base has **no exponentially small ladder**, so it is well-conditioned and the expansion below is well-posed (the certificate logs this).

The expansion. Partition both kernels at the seam site: $K = \begin{pmatrix} A & b \\ b^{\mathsf{T}} & c \end{pmatrix}$, seam pivot $d_0 = c - b^{\mathsf{T}}A^{-1}b$. Along $K(t) = K_{\mathrm{Cram}} + t\,\Delta K$, with $u = \bar{A}^{-1}\bar{b}$, $r = \bar{A}^{-1}\delta b$, $m = \bar{A}^{-1}(\delta A\, u)$ (bars: Cramér; deltas: fluctuation), the resolvent series gives $d_0(t) = \varphi_0 + \varphi_1 t + \varphi_2 t^2 + \varphi_3 t^3 + O(t^4)$ with

$$\varphi_0 = \bar{c} - \bar{b}^{\mathsf{T}}u, \qquad \varphi_1 = \delta c - 2\,\delta b^{\mathsf{T}}u + u^{\mathsf{T}}\delta A\, u, \qquad \varphi_2 = -(r^{\mathsf{T}}\delta b - 2\, r^{\mathsf{T}}\delta A\, u + m^{\mathsf{T}}\delta A\, u),$$

$$\varphi_3 = -2\, r^{\mathsf{T}}\delta A\, m + r^{\mathsf{T}}\delta A\, r + m^{\mathsf{T}}\delta A\, m,$$

computable with one LU factorization of $\bar{A}$ and three solves, exactly, in mp. The reading: φ_1 is the seam's **linear** sensitivity to the zero-oscillation content (its per-prime split shows *which shifts the seam reads*); φ_2 is **pair-correlation-grade** (two fluctuation insertions); φ_3 is **the cubic drift** — three insertions of the arithmetic fluctuation against the seam's conditional geometry: the bridge object. In G.1′'s language, at a marginal contact the approach mode is the seam's innovation direction, ν is the seam-pivot functional itself ($d_0 = \langle \nu, \Omega\rangle$ — the boundary distance *is* the pivot), and the windowed push coefficients at orders $1, 2, 3$ in the fluctuation are exactly $\varphi_1, \varphi_2, \varphi_3$ evaluated along the contact asymptotics.

Diagnostic honesty. If the cubic Taylor fails to capture $d_0(1) - d_0(0)$ (registered G-R1), the failure mode is itself a finding: the exp-small ladder would then be **non-perturbative in the prime fluctuation** — the

zeros' imprint not reachable at any finite fluctuation order — which would relocate the exclusion's obligation to a resummed object and be reported as such, not papered over.

5. Proposition G.2 (the identification) and Theorem G.3 (the conditional exclusion)

Proposition G.2 (the cubic drift is the program's order-3 invariant) [A**; existence/stability measured, §6**]**.** *The windowed cubic drift φ_3, evaluated along the seam's contact asymptotics, is a doubling-type trilinear prime functional — three fluctuation insertions, pairing frequency content at $(0,\gamma,2\gamma)$-type triples through the seam's conditional resolvent — and its non-vanishing welds to the program's unconditional order-3 inventory: the $(0,\gamma,2\gamma)$ Bochner minors (P∞-C Prop. C.3), the Jacobi/Hasse–Davenport chirality (P∞-B Thm. B.2: non-real phases in every admissible pair), the cumulant transfer $\tau_3 = -1$ (N5 Thm. 6.2), and the chiral channel's sign $L(\frac{1}{2},\chi_3) > 0$ ($c_2 < 0$, diez §2.6).* Status: the *existence, sign-stability, non-decoupling, and anatomy* of φ_3 are the certificate's content (§6); the analytic evaluation of φ_3 as a doubling sum — the weld proper — is this paper's named open lemma. It is an order-3 obligation, which is the point: it is **not** pre-killed by the program's order-2 no-go theorems, the first attack in the inventory with that property.

Theorem G.3 (conditional exclusion of marginal contact) [P**-structure | G.2**]**.** *Assume G.2's conclusion: the windowed cubic drift along any boundary-marginal approach of the seam renewal is bounded away from zero. Then no asymptotically stationary marginal boundary contact exists (G.1(iii)/G.1′): the seam orbit either exits the cone at finite radius (RH false) or remains uniformly interior at its own scale (room). Consequences: (a) the marginal corridor between the two alternatives is empty — if RH is true it is true with room, and the orbit-version of the degeneracy face is discharged; (b) the remainder exponent p stabilizes — the lawful deceleration terminates at an interior profile; (c) F.25's contraction operates on the interior side, its (U)-hypothesis now needed only in the form the measured confinement coordinates (F §24) already address.* Caveat carried from §2: the wandering variant is excluded only given orbit convergence (measured: the correlation orbit settles; not proven), and the exit scenario is untouched — the theorem narrows the failure geometry, it does not close RH.

6. The certificate: the first round fires into a discovery (Proposition G.4)

Certificates: `probe_pinfG_cubic_drift.py` (G-R0 GREEN; G-R1, G-R3, G-R5 fired = one diagnosis; G-R4 logged), `probe_pinfG_cramer_indefinite.py` (the diagnosis made spectral); logs and npz adjacent.

6.1 The run, and the lineage

Five windows $\lambda = 1.8$–2.5 at dps 50. **(G-R0)** the instrument's t=1 endpoint reproduces the certified seam remainders to five digits at every window — the lineage to F §23's objects is exact. The expansion record:

λ	d_0^{true}	d_0^{Cram}	gap	φ_1	φ_2	φ_3	recon. resid
1.8	0.694	2.039	−1.345	+0.012	−0.390	+0.173	0.847
2.0	0.547	3.448	−2.901	+0.970	+0.317	+0.914	1.759
2.2	0.375	2.587	−2.212	−0.069	−0.294	−0.022	0.826
2.35	0.315	2.494	−2.180	−0.025	−0.547	−0.029	0.724
2.5	0.256	2.416	−2.160	+0.123	−0.731	+0.141	0.784

(G-R1) fired hard (residuals 0.72–1.76: the cubic Taylor does not capture the gap), **(G-R3) fired** (the φ_3 sign flips $+,+,-,-,+$), **(G-R5) fired** (the longest shifts carry $0.18 < 0.30$ of the linear mass). One diagnosis explains all three.

6.2 The diagnosis, and Proposition G.4

The smoking number is in the table: $d_0^{\mathrm{Cram}}(\lambda{=}2.0) = 3.448 > \mathrm{bare}_0 \approx 2.78$**.** A Schur complement cannot exceed its own diagonal entry on a positive semidefinite form ($d_0 = c - b^{\mathsf{T}}A^{-1}b \le c$ requires $A \succeq 0$); the inequality is therefore a *certificate of indefiniteness* of the Cramér interior. The expansion base sits **outside the positive cone**, the Taylor path $t \in [0,1]$ crosses the cone boundary, and the low-order coefficients are boundary-crossing artifacts, not drift coefficients — G-R3's sign flips and G-R5's anatomy are downstream of the same kill. The registered language anticipated the dichotomy ("no stable cubic drift *at this instrument*"); the corpses are retained.

Proposition G.4 (the fluctuation-free window form is indefinite) [P**; Schur certificate + spectral table, both conventions**]**.** *The PNT-mean window form — the certified builder with its prime atoms replaced by the smooth density $\int e^{u/2}\varphi$ — is not positive semidefinite at any certified window; equivalently, window Weil positivity is carried by the prime fluctuation, not by the smooth background.* The Schur certificate at $\lambda = 2.0$ is unconditional arithmetic of the computed matrix; the spectral table settles it uniformly: minimum eigenvalue $-0.968, -1.039, -1.088, -1.115, -1.138$ across $\lambda = 1.8$–2.5 (negative count $2 \to 4$) under the from-0 convention, and $-0.212 \to -0.322$ (two negative directions) with the fictitious $(0, \log 2)$ mass removed — **indefinite under both conventions, with the violation growing as the window grows.** The smooth world does not drift toward the cone; it drifts away from it. Anatomy of the negative directions (certified, $\lambda = 2.5$): the violations are **low-frequency** — the deepest (-1.138) is the *even, zero-frequency* mode (dominant DFT mode $k = 0$), the next two (-0.690 odd, -0.306 even) sit at $\omega \approx 3.4$ — all entirely inside the zero-free band $\omega < \gamma_1 = 14.13$. The mean field fails exactly where the true form's positivity is carried by the first zero's reach (F.4's floor $= 2\theta'(\gamma_1)$): *what the arithmetic rescues is the sub-γ_1 band, in both parities* — the input to round 2's design.

The meaning, in the program's lenses. This is the geometric-side twin of F.6's zero-side identity (positivity = a sum of squares *over the zeros* — remove the zero-oscillations and there is nothing left to be positive); it is §18.2's conspiracy made global (the prime term does not perturb positivity, it *constitutes* it, jointly with arch and pole); and it is a new sharp instance of the program's order-blindness family (CBS, Tower, P∞-B Thm B.1): **the mean-field prime data — the zeroth-order description of the primes — does not even lie in the positive cone.** The measured gap trend says more: the fluctuation's share of the seam's predictability *grows* with the window ($\mathrm{gap}/d_0^{\mathrm{true}} = -1.9 \to -8.4$ across $\lambda = 1.8 \to 2.5$) — the seam is increasingly arithmetic. Scope guard: the object measured is the *mean field* (fluctuations deleted), not Cramér's random model (which fluctuates); no claim about the latter is made.

6.3 The corrected instrument (round 2, designed)

The bridge must be re-based **inside the cone**: expand around the *true* kernel at window λ, in the direction of the **per-arrival fluctuation**

$$\delta K_{\mathrm{arr}} \;=\; \big[K_{\mathrm{true}}(\lambda') - K_{\mathrm{true}}(\lambda)\big]_{\mathrm{aligned}} \;-\; \big[K_{\mathrm{Cram}}(\lambda') - K_{\mathrm{Cram}}(\lambda)\big]_{\mathrm{aligned}}, \qquad \lambda'^2 = \lambda^2 + 1,$$

the window-growth increment's deviation from its PNT mean — small (one arrival's worth), physical (the orbit's actual driving), and exactly G.1′'s push channel. The same φ-formulas apply with base $= K_{\mathrm{true}}$; costs: the cross-window alignment (the matched-bandwidth locality certified at F §23.1/P2) and F64-6 dps discipline (the true interior's ladder: dps ≥ 65 at $\lambda = 2.5$). Companion instrument: the **diagonal-triple attribution** of φ_3 — the $n_1{=}n_2{=}n_3$ diagonal of the triple sum is an explicit unconditional $\sum \Lambda(n)^3 n^{-3/2}$-grade term; measuring its share decides whether G.2's non-vanishing can rest on unconditional ground.

7. Status and honesty

Stands [P]**:** Lemma G.1 and the Littlewood remark (the exclusion's structural half, consuming only unconditional inputs: cone preservation + $\Omega_{\pm}$); the φ-identities (§4, exact); Proposition G.4 (the discovery — the

smooth world is outside the cone). **Open** [A]**:** G.2's identification, now with its base-point gap *located* — the cubic drift must be measured along the orbit (per-arrival instrument, §6.3), not from the Cramér base; the diagonal-triple road named. **Theorem G.3** awaits G.2 — unchanged in strength, with its falsifiable prediction standing: if the exclusion is real, the remainder exponent p stabilizes ($\lambda = 3.5$ tests it). **Fired and retained:** G-R1/R3/R5 (the base-point kill); the first-draft framing repaired by the §2 audit before any computation. **The route's honest state:** the consumption attempt did not land the cubic tonight; it landed the *reason the naive bridge cannot work* — positivity has no smooth background to perturb around — which simultaneously upgrades the discovery ledger (G.4) and pins the corrected instrument. The conserved core is conserved; the marginal-contact corridor remains the target; the structural half is in place and waiting for its coefficient.

Anchors (§§6–7). `probe_pinfG_cubic_drift.py` + log + `pinfG_cubic_drift.npz`; `probe_pinfG_cramer_indefin` + log + `pinfG_cramer_spectrum.npz`; F.6 (zero-side sum of squares); §18.2/F.19 (the conspiracy; pole-matching); F §23.1 (the alignment machinery for round 2); F64-6 (the dps discipline); P∞-B Thm B.1/CBS/Tower (the order-blindness family G.4 extends); GPD (three falsifiers fired, one diagnosis, corpses kept). *(Rounds 2 and the sum-of-squares assembly: §§8–9, June 13.)*

8. Round 2: the per-arrival instrument inside the cone — non-perturbativity measured, Proposition G.4 localized to one arrival, the directional detection, and the gate's honest verdict

Certificates: `probe_pinfG_cubic_drift_v2.py` + `pinfG_cubic_drift_v2_run.log` + `pinfG_cubic_drift_v2.npz` (mp dps 65; W0a/W0b/W0c GREEN; the gate run at $\lambda \in \{2.0, 2.5\}$, three off-line amplitudes + Frobenius-matched on-line control; lineage anchor: the certified $\lambda = 2.5$ window rebuilt at dps 65 reproduces $R_{\text{seam}} = 0.09216$ exactly). Run date June 13, 2026.

8.1 The corrected frame, and the two smoke discoveries

The §6.3 design was realized on the λ'-grid ($\lambda'^2 = \lambda^2 + 1$), where the builder's linearity in the prime measure makes the alignment exact: base = the true kernel with one band's arrivals replaced by their PNT mean, direction $\delta K_{\text{arr}} = K_S - K_A$ (arch/pole/$W_{\mathbb{R}}$ cancel identically), endpoint $t{=}1$ = the certified true kernel. Two discoveries surfaced **at the smoke stage** and were locked into the design before the run:

(SMK-1, the boundary node.) An atom at the window edge $y = L'$ is *invisible in its own frame*: $\sin(2\pi k\, y/L') = 0$ for every k and the Fejér ramp vanishes there — arrivals phase in continuously as the window grows past them; there is no impulsive arrival. (The first-design band $(\lambda^2, \lambda^2{+}1]$ placed the $\lambda{=}2.0$ atom $n{=}5$ exactly on the node: its pure-atom cubic measured 10^{-78}.) The per-arrival push is therefore read at the **crossover band** $(\lambda^2 - 1, \lambda^2]$ — one rung behind the edge, exactly where the saturation wave (F.22/§20) carries the eating. Band atoms: $n = 3, 4, 4, 5$ at $\lambda = 1.8, 2.0, 2.2, 2.35$ and **none at** $\lambda = 2.5$ ($n = 6$ composite — the pure-smooth control step, free).

(SMK-2, the radius.) Replacing *one band's* arrivals by their mean already exits the cone (§8.3) — so the Taylor path from the mixed base to the true kernel re-crosses the boundary, and round 1's lesson recurs one level deeper: there is **no Taylor-accessible interior segment of one-arrival length in the fluctuation direction.** The φ's are **germ objects** at the true kernel — which is precisely what Lemma G.1 consumes (DT, D^2T, D^3T *at the point*); what dies is only the demand that the order-3 polynomial capture the full step. Validity is certified by an exact algebra unit test (φ-formulas vs Richardson differences on a random exact instance: $5\times10^{-44}, 1\times10^{-40}, 5\times10^{-22}$ — which also retro-guards round 1's formulas) plus the radius readings below; the pole-normalized **shape invariant** $g_3 = \varphi_3\varphi_1/\varphi_2^2$ ($\equiv 1$ for a pure single-pole resonance) joins the raw φ_3 as the carrier of cubic content.

8.2 Proposition G.4′ (the per-arrival localization of G.4) [P; certified at five windows]

Replacing the arrivals of a single unit-ψ-mass band by their PNT mean makes the window form indefinite at every certified window: float min-eig $-0.26, -0.30, -0.20, -0.23, -0.28$ at $\lambda = 1.8$–2.5 (mp Cholesky fails; the true bases all pass at dps 65) — **a violation of window-independent depth ≈ -0.25 from one arrival's worth of smoothing** (band ψ-fluctuations between $+0.03$ and -0.42, both signs). Positivity is *identity-protected per arrival*, not stability-protected: each arrival's exact fluctuation is load-bearing against the exp-small ladder. This sharpens G.4 (the all-smooth world is outside the cone) to single-arrival granularity, and it quantifies §23.5's exponential-sensitivity finding at the arrival scale. Consequence for instruments: the mixed-base endpoint values $d(0)$ are Schur data of an indefinite form — **not program-grade observables** (F64-5); the registered S6 interiority reading is accordingly demoted to indicative.

8.3 The measured anatomy: the radius law, and single-pole domination

True-base germ data, d_- channel (the chiral block; the direction δK is reflection-invariant to 10^{-64}, so this is not §3's symmetry-killed channel):

λ	band atom	ψ-fluc	d_-(true)	φ_1	φ_3	g_3	$t_0 = \lvert\varphi_1/\varphi_2\rvert$
1.8	3	+0.028	3.04e−3	−1.71e6	−7.49e21	0.9979	1.5e−8
2.0	4	−0.189	4.49e−4	−1.22e7	−1.67e26	0.9950	2.7e−10
2.2	4	−0.134	4.45e−5	+2.99e7	+1.53e29	0.9873	1.4e−11
2.35	5	+0.273	4.73e−6	−8.41e7	−2.40e32	0.9872	5.9e−13
2.5	—	−0.417	6.51e−7	−1.93e8	−1.23e35	0.9854	3.9e−14

(i) The non-perturbativity is measured, with its law. The germ's convergence radius t_0 falls $1.5 \times 10^{-8} \to 3.9 \times 10^{-14}$ — the per-arrival fluctuation sits 10^8–10^{14} *outside* the germ's disc, confirming §4's anticipated branch: *the exp-small ladder is non-perturbative in the prime fluctuation; the zeros' imprint is not reachable at any finite fluctuation order; the exclusion's obligation relocates to a resummed object.* Measured pace: t_0 falls at roughly 0.35–0.4 of the ladder's Fuchs pace across the sweep (a reading, mechanism open). **(ii) Single-pole domination.** $|\varphi_2/\varphi_1| \approx |\varphi_3/\varphi_2|$ at every window and $g_3 = 0.985$–0.998: the germ is a near-pure one-pole resonance — the cubic's raw magnitude is resolvent amplification, with the genuine shape content confined to the small deviation $g_3 - 1$. Consequently the registered S1/S2 rules auto-survive **vacuously** ($\varphi_3/\varphi_1 = 1/t_0^2 > 0$ trivially; the ratios *rise* at Fuchs pace), exactly as the registration's caveat anticipated; they are not evidence for a stable arithmetic cubic. **(iii) The attribution is structure-fragile (S3).** The pure-atom triple share of φ_3 is ≈ 0 at $\lambda = 1.8, 2.0, 2.35$ (the smooth-deficit insertions dominate) and 0.22 at $\lambda = 2.2$ with sign-alternating ladder — rung-level resonance anatomy, not invariant-grade data (F64-5 in action: only the total φ_3 is determinant-ratio-grade; its decomposition is not). **The diagonal-triple road to unconditional ground has no empirical support at this instrument.**

8.4 The gate: the verdicts, and the named outcome

Deformation family: $\mathrm{d}\rho(u) = -4[\cosh(\delta u) - 1]\cos(\gamma_1 u)$ (the FE-symmetric off-line quadruple $\frac{1}{2} \pm \delta \pm i\gamma_1$ in the prime-side density), $\delta \in \{0.1, 0.2, 0.3\}$; on-line control = the pair moved $\gamma_1 \to \gamma_1 + 1$, Frobenius-matched to $\delta = 0.2$. Verdicts, in decreasing sharpness:

(a) The directional detection [P-grade datum]. The off-line family breaks positivity at **every** amplitude tested — already at $\delta = 0.1$, Frobenius 1.5×10^{-2} — while the on-line control **preserves** positivity at $4\times$ larger Frobenius (6×10^{-2} at λ=2.0, 1.1×10^{-1} at λ=2.5), at both windows. **The cone's thin direction *is* the off-line direction**: bounded-window positivity is tolerant of on-line rearrangement and intolerant of off-line displacement — a sharp, certified instance of the order-2 detection channel (§23.5's two-sided law, now directional).

(b) The raw cubic's response is pure ladder (the leak, measured). Every deformation — off-line *and* the PSD-preserving control — collapses the deformed φ_3 by 11–14 orders (e.g. λ=2.0: $-1.67\times10^{26}\to10^{12}$–$10^{19}$), so the response $\approx -\varphi_3^{\text{base}}$ identically and $R_{\text{spec}}=1.00$ at both windows: **the raw cubic reads deformations only through the order-2 resolvent/ladder rearrangement.** (The off-line collapse is 5–6 orders deeper than the control's — itself a directional signature, but order-2-explicable via the cone exit.)

(c) The shape cubic's response is window-fragile. g_3 responds off-line-specifically at $\lambda=2.0$ ($|\Delta g_3|=0.35$–1.27 vs control 7×10^{-4}: $R_{\text{spec}}=495$) but **non-monotonically in** δ, and at $\lambda=2.5$ it barely responds at all (5×10^{-4} vs control 3.5×10^{-4}: $R_{\text{spec}}=1.5$). The $\lambda=2.0$ signal is consistent with pole-competition in the *indefinite* deformed interior (an order-2 mechanism switched on by the cone exit), and its non-uniformity across windows forbids reading it as a stable chirality signature. Two instrument flaws acknowledged for the record: the measured dps-jitter J came out 0.00 — an artifact of float-casting mp values agreeing beyond float64 resolution — so GATE-A's formal threshold ($10J$) could never fire and the operative noise reference is the on-line control; and the S5 console line prints $p/2$ (the fit and predictions are correct).

(d) The named outcome. The §2 audit said the RH-false world contains the same primes; the gate completes this into a measured statement: **at bounded window there is no interior off-line regime to test against** — zero displacement manifests as immediate order-2 cone exit (item a), and every response channel of the cubic instrument is exhausted by that mechanism (items b, c). The per-window φ_3 is unconditional bounded-window data; it cannot, by itself, consume an RH-true input. **The cubic route's entire conditional content therefore lives in one asymptotic inequality** — *the windowed cubic drift along any boundary-marginal approach of the seam renewal stays bounded away from zero as* $\lambda\to\infty$ (G.2's weld, now with its finite-window approaches provably uninformative) — supportable but not certifiable by finite windows. This is reported plainly as the round's principal negative: **G.2 remains without empirical support, and no finite-window deformation instrument of this class can supply it.** The structural half (G.1 + Littlewood) and the falsifiable prediction of G.3 (p stabilizes; $\lambda=3.5$) stand unchanged. *(Referee note, post-hoc lens: this outcome is the* ***Kummer-rigidity of the index character****. In the quanta-object frame (L., working note* `quanta.md`*, June 2026, §4) the order-3 sign is the index's gauge-rigid character — summoned by the index, present for every dataset and every gauge — while only the gauge-fluid radicand support carries world-dependence; the gate's* $R_{\text{spec}}=1.00$ *is that rigidity, measured. The verdict this round established empirically is thereby derivable a priori from Kummer theory: a character cannot discriminate worlds; only support can, and support is exactly the per-arrival drift the two-regime argument places beyond finite windows.)*

8.5 The positive readings

(S5) Five new ladder points. R_{seam} at the λ' windows $(2.059, 2.236, 2.417, 2.554, 2.693)$ = $0.1744, 0.1542, 0.1030, 0.0932, 0.0819$ — within $-4.2\%/+10.2\%/-5.7\%/+1.9\%/+6.0\%$ of the certified $p\approx3.2$ power law, the largest deviation at $\lambda'=\sqrt{5}$ adjacent to the logged arrival-phase wiggle: **the RG-time decode extends to five interleaved windows from an independent build family. (W0 suite)** true-base PSD at dps 65 everywhere (F64-6 honored); reflection defects $\le 1.4\times10^{-64}$; the t=1 lineage exact. **(G.4′)** is itself the round's strongest positive deposit, with (a)'s directional detection beside it.

8.6 Status after round 2

The consumption attempt's measured face is now complete at certified windows, and the verdict is honest: *the cubic's finite-window values are resolvent-dominated, unconditional, and deformation-blind beyond order-2 mechanisms* — the conditional load of the exclusion is exactly one asymptotic inequality, named in §8.4(d). What the round deposits: G.4′ (per-arrival identity-protection of positivity), the directional detection (the cone's thin direction = off-line), the non-perturbativity law (t_0 at a third of the Fuchs pace), the single-pole

anatomy (g_3 tables), five power-law ladder points, and the instrument discipline (germ objects + algebra unit test + crossover band).

The redirect (L.). The first-draft route-forward — "the resummed object as the cubic's next instrument" — repeats the category error one level up: the resummed $d(t)$, read at finite window, is *still* RH-unconditional (the two-regime argument applies to it verbatim), so it cannot serve discrimination, and the cubic was a proxy for discrimination. **Drop the proxy.** The resummed object's actual value is that it gives the *exact per-arrival increment* — the literal content of the open step ("the increment stays bounded below the remainder, forever" = (U) directly) — and that is where Theorem G.5 earns its keep: the square-positive gearing is the sign-bookkeeping that routes the increment's sign content through the one chiral scalar. The increment-boundedness line never needed discrimination, only a uniform asymptotic bound. The instrument this prescribes — exact per-arrival increments of the certified observables across true-to-true unit-λ^2 pairs, the G.5 gearing split, the channel-resolved (U)-band, and the arrival-direction cone margin t^* — is executed in §10.

Anchors (§8). §§2–6 (audit, G.1+Littlewood, the φ-formulas, G.4, the §6.3 design); F §§20, 23.5, 23.6 (crossover band; exponential sensitivity; the power-law decode); F64-3/5/6 (near-radical directions; observable discipline; dps floors); Littlewood 1914; GPD (SMK-1/SMK-2 locked pre-run; S1/S2 auto-survival flagged; jitter and S5-print flaws recorded); `probe_pinfG_cubic_drift_v2.py`, run log, npz.

9. Strike 2: the sum-of-squares assembly — the square/Möbius bridge (Theorem G.5), the chiral-sign certificate, and the CC-2021 weld

Certificate: `probe_pinfG_sos_bridge.py` + `pinfG_sos_bridge_run.log` (B1 GREEN: the closed forms hold on all six certified `parity_totals` windows at storage grade, $\le 10^{-12}$ relative on the float lift; B2 GREEN: full mp rebuild of the $\lambda = 2.0$ window through the certified parity machinery, residuals 5.0×10^{-42} and 1.7×10^{-41} at dps 50).

9.1 The frame: the canonical SOS, and where the sign obligation lives

By Theorem F.21 the window Weil form *is* its own canonical sum of squares: the Gohberg–Krein chain factorization gives $QW_\lambda(f) = \sum_m d_m\,(\ell_m f)^2$ over the center-out cells, and by F.27(a) the cells split into two independent parity towers, so **RH $\iff$ the canonical SOS never needs a negative square, in either sector.** The published shape this assembles toward is Connes–Consani 2021, Theorem 1: the geometric Weil functional dominates a trace of manifest squares, $W_\infty(g \star g^*) \;\ge\; \mathrm{Tr}(\vartheta(g)\, S\, \vartheta(g)^*) \;\ge\; 0$ — an archimedean positive-operator exhibition whose defect is the prolate crawl μ_λ. The chain SOS is the *exact finite-window completion* of that exhibition (equality, not inequality; every weight a finite computation, permanent by the martingale immunity). The assembly question of this strike: **in the per-arrival refresh of the canonical weights, which single scalar carries the sign obligation?** The answer is the bridge identity's exact form.

9.2 Theorem G.5 (the square/Möbius bridge) [P; **certified**]

With the F.27(b) frame — sector edge pivots $d_\pm$ over the reflection-closed interior, $v_0 = \mathrm{Var}(\delta_0 \,|\, \mathrm{int}')$, $\gamma = \mathrm{Cov}(\delta_0, E_1 \,|\, \mathrm{int}')$, $G = \gamma/\sqrt{2}$, $g = G/d_+$, seam pivot $d_0 = v_0 - \gamma^2/d_+$ — the certified mixed-edge pivot obeys, exactly:

$$\textbf{(G.5a)}\quad d_{\mathrm{std}} \;=\; \frac{v_0}{2} + d_+ + 2G - \frac{(d_+ + G)^2}{d_+ + d_-}, \qquad \textbf{(G.5b)}\quad d_{\mathrm{std}} - \frac{d_0}{2} \;=\; \frac{(d_+ + G)^2\, d_-}{d_+\,(d_+ + d_-)},$$

$$\textbf{(G.5c)}\quad \frac{\partial\, d_{\mathrm{std}}}{\partial d_-} \;=\; \Big(\tfrac{1}{2} + Y\Big)^2, \qquad Y = \frac{(d_+ - d_-)/2 + G}{d_+ + d_-}\;:$$

the d_--derivative of the certified pivot is a perfect square everywhere, and the excess over the half-seam is the chiral pivot times a square over the sector-positive denominator $d_+(d_+ + d_-)$.

Proof. Substitute $s = d_+ + d_-$ in F.27(b); with $(d_+ - d_-)/2 + G = (d_+ + G) - s/2$ the square expands as $(d_+ + G)^2 - (d_+ + G)s + s^2/4$, the $s/4$ terms cancel, and (G.5a) falls out; (G.5b) is (G.5a) minus $d_0/2 = v_0/2 - G^2/d_+$; (G.5c) is the x-derivative of (G.5a), recognizing $1/4 + Y + Y^2 = (1/2 + Y)^2$. □

Retro-correction of F.27(c). The first-draft "$O(d_-^2)$" tail is exactly the expansion of the Möbius denominator $1/(d_+ + d_-)$ — the additive decomposition is *closed-form*, with $\kappa_{\text{exact}} = (d_+ + G)^2/(d_+(d_+ + d_-))$ replacing the constant $(1+g)^2$ (they agree at $d_- = 0$). Checked against the certified archive: the constant-κ approximant $(d_0/2 + (1+g)^2 d_-)/d_{\text{std}}$ overshoots by 29–42% at $\lambda \geq 1.8$ — the inverse of the Möbius factor $d_+/(d_+ + d_-) \approx 0.41$–$0.50$ on the odd term — while the exact form closes the identity to 1.000000000 at every window (storage grade). Referee note landed in P∞-F §24.2.

9.3 Corollaries: the chiral-sign certificate, and the measured interiority of the chiral channel

Corollary G.5.1 (the chiral-sign certificate) [P]. *If $d_+ > 0$ and $d_+ + d_- > 0$ (both certified per window, permanent), then*

$$\operatorname{sign}\Big(d_{\text{std}} - \tfrac{d_0}{2}\Big) \;=\; \operatorname{sign}(d_-) \qquad \text{unless } g = -1 \text{ (where both sides vanish):}$$

the certified mixed ledger — the carrier of the published remainder law — reads the chiral sector's sign with no odd-chain computation. The degenerate locus $g = -1$ is exactly the measured $\lambda = 1.5$ κ-transient of F.27(iii), retro-explained: the bridge loses its bite precisely where the data said it did.

Corollary G.5.2 (measured chiral interiority). In G.5-coordinates the certified oddshare $= (d_{\text{std}} - d_0/2)/d_{\text{std}} \in [0.24, 0.30]$ ($\lambda \geq 1.8$, F §24.3) reads: **the chiral pivot's certified share of the mixed ledger is bounded away from 0 and 1 at every window** — direct (U)-interiority data for the chiral coordinate. Registered consequence: *marginal boundary contact of the chiral channel requires* oddshare $\to 0$; the $\lambda = 3.5$ extension tests the plateau.

9.4 The per-arrival assembly, and the weld

Writing $E = d_{\text{std}} - d_0/2 = \tilde{\kappa}\, d_-$ with $\tilde{\kappa} = (d_+ + G)^2/(d_+(d_+ + d_-)) \geq 0$, the per-arrival increment is, exactly,

$$\Delta E \;=\; \tilde{\kappa}_{n+1}\, \Delta d_- \;+\; d_{-,n}\, \Delta\tilde{\kappa},$$

with $\tilde{\kappa}$ a manifest square over a sector-positive denominator and $\Delta\tilde{\kappa}$ built from even-sector motion only. **Given the even tower** (the cosh/ζ channel — the CCM-covered side: prolate positivity proven at every finite stage), every coefficient in the refresh of the certified excess is a square or a sector-positive Möbius factor, and **the only sign-critical scalar per arrival is the chiral refresh Δd_- — strike 1's measured object.** This is the sum-of-squares assembly in its honest form: not a positivity proof, but an exact localization — the canonical SOS needs a new non-negative square once per arrival, in one named channel, and the program's unconditional order-3 inventory ($c_2 < 0$ via $L(\frac{1}{2}, \chi_3) > 0$; the Jacobi phases; $\tau_3 = -1$) lives in exactly that channel.

What is and is not closed. Closed [P]: G.5(a–c), G.5.1, and the increment identity — exact, certified, permanent. Measured: G.5.2's plateau. Open, unchanged: even-tower positivity forever (the CCM crawl) and the chiral refresh's sign forever (= (U) split by parity, F.27(a)); the weld of strike 1's φ_3 to the chiral inventory (G.2). The deposit: RH's per-arrival sign obligation now has an exact two-channel ledger in certified coordinates, with the odd channel's obligation carried by one scalar whose drift is the cubic instrument's target — strikes 1 and 2 meet at Δd_-.

Anchors (§9). F.27(b–c) (the bridge; the probe's κ-table deficit = the Möbius factor); F.21/F.23 (chain SOS; Levinson); Connes–Consani 2021 (*Weil positivity and trace formula*, Thm 1 — the positive-operator shape; CCM 2511.22755 for the convergence form); N5 Thm 6.2 / diez §2.6 / P∞-B Thm B.2 (the chiral inventory); F §24.3 (oddshare plateau, ρ bands); `probe_pinfG_sos_bridge.py`, `pinfG_sos_bridge_run.log`.

10. Round 3: the increment-bound instrument — the (U)-band measured at five exact arrival steps, the gearing's compensation law, and the arrival-direction margin

Certificates: `probe_pinfG_increment_bound.py` + log + `pinfG_increment_bound.npz` (dps 65; L1 lineage exact to the bit, N1 4×10^{-42}); `probe_pinfG_margin_law.py` + log + `pinfG_margin_law.npz` (the N6 follow-up). This section executes §8.6's redirect (L.): the resummed object is used **for the increment, not for discrimination**.

10.1 The instrument

The §8 run built true kernels at $\lambda'^2 = \lambda^2 + 1$, so the program now owns **five exact unit-λ^2 pairs with both endpoints inside the cone**: $(1.8 \to 2.0591)$, $(2.0 \to 2.2361)$, $(2.2 \to 2.4166)$, $(2.35 \to 2.5539)$, $(2.5 \to 2.6926)$ — true-to-true, no mixed base anywhere. Per pair, the probe computes the exact per-arrival increments of the certified bridge observables $(d_0, d_-, d_+, E = d_{\rm std} - d_0/2)$, the G.5 gearing split, the channel eating fractions, and (3b) the cone margin along the arrival direction. Instrument verdicts: **L1** — the rebuilt λ'-window pivots reproduce the §8 archive *exactly* (0.0 deviation: same code path, deterministic); **N1** — $E = \tilde{\kappa} d_-$ at $\leq 4 \times 10^{-42}$ at every new window; **N1b** — the Leibniz split $\Delta E = \tilde{\kappa}' \Delta d_- + d_- \Delta\tilde{\kappa}$ verified at $\leq 5 \times 10^{-15}$ on the four mp-certified pairs; the registered 10^{-10} rule *fired* on the 1.8-pair at 1.7×10^{-9} — diagnosed as the storage grade of the $\lambda = 1.8$ endpoint (a float-lift window in `parity_totals`), not an algebra error; threshold artifact, retained.

10.2 The gearing's compensation law (measured)

In log coordinates the G.5 identity reads $\Delta \ln E = \Delta \ln \tilde{\kappa} + \Delta \ln d_-$ per arrival, and the five pairs give, exactly (identity to the third decimal at every pair):

pair	$\Delta \ln \tilde{\kappa}$ (gear)	$\Delta \ln d_-$ (chiral)	$\Delta \ln E$ (residue)
$1.8 \to 2.059$	$+2.435$	-2.764	-0.329
$2.0 \to 2.236$	$+2.305$	-2.575	-0.269
$2.2 \to 2.417$	$+2.183$	-2.559	-0.377
$2.35 \to 2.554$	$+2.627$	-3.099	-0.472
$2.5 \to 2.693$	$+2.589$	-2.896	-0.308

In raw terms the two split terms ($\tilde{\kappa}' \Delta d_- \approx -1.30 \ldots - 0.50$ vs $d_- \Delta\tilde{\kappa} \approx +1.27 \ldots + 0.49$) cancel to 2.5–4% of their size. **The mechanism behind F.27(iii) is now measured per arrival: the chiral channel's massive eating ($-c_- \approx -2.8$) is geared down by the square coefficient's growth ($+2.2$ to $+2.6$) into exactly the seam-class marginal slide of the certified excess.** The sign-bookkeeping is exact and certified; the only sign-carrying scalar is d_- (G.5.1), whose per-arrival eating the table now quantifies.

10.3 The (U)-band, channel-resolved at five phases

Eating fractions $f = 1 - d(\lambda')/d(\lambda)$ and margins: **seam** $f_0 = 0.313, 0.239, 0.245, 0.266, 0.243$ — margin $1 - f_0 \geq 0.69$ at every phase, stable-to-growing (the predicted $p/(2\lambda^2)$ slide); **massive cells** $f_\pm = 0.92$–0.96, i.e. $c_- = 2.76, 2.57, 2.56, 3.10, 2.90$ and $c_+ = 2.60, 2.99, 2.71, 3.27, 3.21$ — inside the certified band at all interleaved phases (N3 survives); **the excess** $f_E = 0.24$–0.38. **N2 (G.5.2's first test): the oddshare plateau extends** — $0.2687, 0.2873, 0.2770, 0.2418, 0.2334$ at the five new windows, all inside the registered $[0.20, 0.34]$. **Watch-items flagged for** $\lambda = 3.5$ (drifts within bands, honestly recorded): oddshare drifts down across the last three phases; d_-/d_+ rises $0.96 \to 1.93$ (beyond the certified $[1.0, 1.5]$ snapshot); the merged ten-window fits give $E \propto \lambda^{-3.73}$ against the seam's $\lambda^{-3.25}$ — the odd bridge runs slightly steeper than the seam, which is the drifts' law-form.

10.4 The densified ladder

Eleven windows (six certified + five new): $\boxed{p = 3.204,\ \mathrm{rms}(\log) = 0.050}$ — the decode's tightest fit; the lone $> 6\%$ outlier is $\lambda' = \sqrt{5}$ ($+10\%$, the logged arrival-phase wiggle). Per-pair seam slopes $(0.503, 0.292, 0.374, 0.192, 0.118)$ wiggle ± 0.15 around $p/(2\bar{\lambda}^2)$ ($0.43 \to 0.24$): the law is the robust object, the per-pair slopes carry the phase noise.

10.5 The arrival-direction cone margin: Proposition G.4″ (the margin law)

Round 3's linear bisection saturated its resolution at **every** window: the kernel loses positivity within $2^{-29} \approx 1.9 \times 10^{-9}$ of the true kernel along the arrival-swap direction — already a sharpening of G.4′ (*one billionth of one arrival's smoothing exits the cone*). The log-bisection follow-up (3b) resolves the law:

$$s^* \;=\; 5.1 \times 10^{-11},\ 1.30 \times 10^{-12},\ 1.34 \times 10^{-12},\ 3.6 \times 10^{-15},\ 2.6 \times 10^{-16} \qquad (\lambda' = 2.06 \to 2.69),$$

$$\log_{10} s^* \;\approx\; -1.73\,\lambda'^2 - 2.9 \quad (\mathrm{rms}\ 0.54),$$

i.e. the margin collapses exponentially at ≈ 0.37 of the Fuchs pace — **the same pace as the §8 germ radius** t_0 (-1.86 per unit λ^2), with t_0 exceeding s^* by 1–2.5 decades pointwise (ordering as it must be: the PSD crossing binds at or before any channel's germ pole; the binding mode is not the odd block's dominant one). The arrival-phase wiggle reappears as the 2.24–2.42 flat. **Reading:** (U) **in arrival units is not carried by a roomy cone — the margin dies exponentially; the renewal survives because the true arrival is exactly the rescued direction (the conspiracy as per-arrival identity), and the** (U)**-relevant interiority lives in the renormalized/Thompson coordinates of F.25, where §10.3's band is the measurement.**

10.6 Status after round 3

The open step's literal quantities — the per-arrival increments against the remainders — now have a measured five-phase, channel-resolved band in certified coordinates, with the G.5 gearing supplying exact sign-bookkeeping and the margin law quantifying why no naive stability argument can close it (the cone margin in arrival units $\to 0$ exponentially; only the renormalized band can stay uniform). The conserved core is conserved: (U) = the §10.3 band holds forever. The $\lambda = 3.5$ discriminator is now **triple-loaded**: the p-stabilization (G.3's prediction), the oddshare plateau (G.5.2), and the §10.3 drift watch-items. Deposits: the compensation law (10.2), the (U)-band (10.3), $p = 3.204$ on eleven windows (10.4), G.4″ (10.5), and the five reusable true-to-true pairs as a standing instrument.

Anchors (§10). §8.6 (the redirect, L.); Thm G.5/G.5.1/G.5.2 (§9); F.25/F.26 ((U), Thompson frame); F.27(iii) (the compensation, now measured); F §23.6 (the decode; the wiggle); Prop G.4′ (§8.2); `probe_pinfG_increment_bound.py` + log + npz; `probe_pinfG_margin_law.py` + log + npz.

11. Round 4: the $\lambda = 3.5$ verdicts — p holds at 2.5%, the plateau decodes as an irrelevant RG direction, and the three-way conspiracy (Proposition G.7)

Certificates: `probe_pinfG_lambda35.py` + log + npz (dps 110, dims 87/91 — both windows PSD, reflection 10^{-109}, d_0-consistency 10^{-57}; the registered U-suite); `probe_pinfF_memory_rate.py` + log + npz (the prime-free reference chain) and the Cramér-pace check (inline, logged). Run date June 13, 2026.

11.1 The p-discriminator: G.3's prediction passes its registered test

$R_{\mathrm{seam}}(3.5) = 0.03422$ vs power-law prediction 0.03336 ($|\Delta \ln| = 0.025$) vs exponential track 0.01185 ($|\Delta \ln| = 1.0$

The RG-time power law extends to $\lambda = 3.5$ **at** 2.5% — the registered F §23.6 discriminator resolves for the power law by a factor $e^{1.04}$, and G.3's falsifiable prediction (*if the exclusion is real, the remainder*

exponent stabilizes) **passes**. Thirteen-window refit: $p = 3.006$, rms(log) $= 0.079$ — the new no-atom step $3.5 \to 3.64$ shows a genuine local inversion (R_{seam} rises to 0.03959), the strongest arrival-phase wiggle yet, pulling the global p from 3.204 to 3.0; the law-class verdict is unambiguous, the third digit of p is phase-limited. The gearing identity continues exactly ($\Delta \ln \tilde{\kappa} + \Delta \ln d_- = \Delta \ln E$ at 3×10^{-16}), and the sixth pair's edge constants soften to $c_- = 2.05$, $c_+ = 1.98$ (below the certified 2.6–3.3; arrival-phase confound noted, class question — pure exponential vs slowly varying — flagged for $\lambda \geq 4$).

11.2 The plateau breaks lawfully: the irrelevant direction (the bands were flows)

The registered U3/U4 rules **fired**: oddshare $= 0.1350, 0.1217$ and $\rho = -0.445, -0.429$ at $\lambda = 3.5, 3.64$ — outside the certified bands $[0.24, 0.30]$ and $[-0.76, -0.60]$. The decode (fits over $\lambda \geq 2.35$, twelve windows):

$$\text{oddshare} \propto \lambda^{-1.87} \text{ (rms 0.037)}, \qquad \rho^2 \propto \lambda^{-1.77} \text{ (rms 0.004)}, \qquad E \propto \lambda^{-5.36} \approx \lambda^{-(p+2)} \text{ (rms 0.036)} :$$

one irrelevant RG direction with τ-eigenvalue $\approx -0.9 \pm 0.1$ (consistent with -1) in $\tau = \ln \lambda^2$. The §10.3/§24.4 "compact bands" were early-time snapshots of this lawful flow — the same decode pattern as §23.6's constant-c (apparent constants = local values of slow power laws), one level up. **The renewal's limit profile is seam-only**: the wrap cell asymptotically decorrelates from the parity edges ($\rho \to 0$), the odd bridge dies one e-fold faster than the seam ($E \sim \lambda^{-(p+2)}$, i.e. oddshare $\sim e^{-\tau}$), and the parity edges ride as slaved passengers. Honest readings: (i) this is *not* a chiral-marginality alarm — $d_- > 0$ at both new windows, falling at its massive pace; what fades is the *certified observable's chiral sensitivity*, polynomially (λ^{-2}, not exp-small) — G.5.1's sign certificate remains usable at polynomial cost; (ii) (U)'s content **concentrates onto the seam cell alone**: uniform interiority of the limit profile = the seam's own law continuing ($p < \infty$) plus lawful irrelevant corrections — the open step's support shrinks to one cell's one exponent, with the approach to the profile now measured. Referee notes landed in P∞-F §23.6 and §24.4.

11.3 Proposition G.7 (the three-way conspiracy; the eating is the fluctuation) [P-grade certificates]

The strike-1a numerics (registered M-suite): **(a) the prime-free window form (arch + pole only) is indefinite at every certified window** (M1 fired as a discovery; mp Cholesky fails $\lambda = 1.8$–2.69, both bandwidth families — not a criterion artifact), and its boundary cells are nearly unsaturated: $R_+ \approx 0.74$–1.3, pace $c_+^{\text{ref}} \approx 0.14$ vs the true 2.81; seam $p_{\text{ref}} = 0.60$ vs the true 3.20. **(b) The PNT-smooth (Cramér) form is likewise unsaturated where readable** ($c_+ \approx 0.08$–0.19 at the late windows; early-window values are G.4 crossing artifacts, flagged). Hence, with G.4 and G.4′: *arch+pole alone — outside the cone; arch+pole+smooth primes — outside the cone; only the full arithmetic object is inside* — **the cone has no smooth interior reference point in any decomposition**, and **the renewal's memory-death (the contraction constant $c_{\text{clean}} \approx 2.7$–$3.8$) is created by the prime fluctuation itself**, not by the archimedean background and not by the PNT mean. Consequence, stated plainly: the effective-contraction strike *as designed* (ground the Bougerol/Birkhoff constant on an unconditional reference chain) is **dead** — F.24(i)'s "the majorant discards the rescue" is total. What survives and sharpens: the contraction is real and measured *on the true kernel*; its constant is an arithmetic-fluctuation quantity; and the cornering reformulation is the **thin-direction program** — the cone's thin direction at window λ is computable (the §8.4 gate's off-line family is its symbol-level model), the orbit's budget along it is the measured margin law $s^*(\lambda)$ (G.4″), and the open step's quantitative face is the boundedness of the thin-direction prime functional window by window, posed on the seam-only limit profile of §11.2.

11.4 Status after round 4

The session's arc closes coherently: the law holds and tightens (p confirmed at 2.5% at the registered discriminator); the boundary cluster's shape flow is decoded (one irrelevant direction, eigenvalue ≈ -1); the open step concentrates onto the seam cell's single exponent; and every scaffold softer than the full arithmetic — Cramér (G.4), one smoothed arrival (G.4′), prime-free (G.7) — is now certified outside the cone, with the

contraction itself certified arithmetic. Queued next (the cornering instruments, in order): the thin-direction stabilization measurement (bottom-subspace convergence across the thirteen windows in the RG frame — if the direction stabilizes, RH = boundedness of one explicit prime-functional sequence with budget $s^*(\lambda)$); the fixed-profile cubic (the §8.4(d) asymptotic inequality, now posed on the *measured seam-only profile* where stationarity quotients out the resolvent scale that killed the window cubic); and the $\lambda \geq 4$ extension (the edge-class question; dps $\gtrsim 140$).

Anchors (§11). F §23.6 (the registered discriminator; the decode pattern); F §24.3–24.4 (the bands now flows — referee notes landed); G.3 (the prediction tested); G.4/G.4′/G.4″/G.5/G.5.1 (the scaffold exclusions; the gearing; the sign certificate's polynomial cost); F.24(i) (the bracket's honest limit, now total); `probe_pinfG_lambda35.py`, `probe_pinfF_memory_rate.py`, logs, npz archives.

12. The §11.4 queue executed: the thin direction is the infrared/prolate sector (stabilized), the profile-cubic seed, and the edge-class softening

Certificates: `probe_pinfG_thin_direction_v2.py` + log + npz (the corrected instrument; v1 retained as a corpse); `probe_pinfG_profile_cubic_seed.py` + log + npz (five new windows, eleven exact pairs); `probe_pinfG_lambda36_pair.py` + log + npz (dps 120). Run date June 13, 2026 (the same night, on Leo's GO).

12.1 Queue (i) — the thin direction: identified and stabilized

The v1 corpse (two lessons, banked). The first instrument read a *single* near-null vector of $K(s^*)$ — F64-5-noncompliant: the crossing modes form a **near-degenerate cluster** (diagnostic at $\lambda' = 2.4166$: two starts converged to distinct near-null vectors, $\theta = 1.2 \times 10^{-20}$ and 9.8×10^{-22}, mutual overlap 1.7×10^{-10}); and the v1 run's θ-values were not reproducible in a clean rerun (matched no near-null mode; superseded — *near-null readings must be cross-checked against an independent eigensolver*, banked as the F64-5 corollary for crossings).

The v2 instrument and the result. Bottom-3 deflated inverse iteration at the measured crossings (residuals $\leq 1.8 \times 10^{-64}$; the bottom θ tiny-*negative* at the bisected s^* — $-3 \times 10^{-18} \to -1 \times 10^{-33}$ across windows — exactly as a bisection midpoint should sit; the deflated rungs match the independent float diagnostic's ladder). The invariant object — the subspace-averaged spectral profile — gives: **(a) stabilization is total**: cross-window overlaps ≥ 0.997 at all five windows; **(b) the identity of the thin direction**: the profile's lobes sit at $\omega = 2\pi k/L'$, $k = 0, 1, 2, 3$, *exactly* ($4.35 = 2\pi/1.445$ at $\lambda' = 2.06$; $3.17 = 2\pi/1.982$ at 2.69), with fixed weights and **cluster parity exactly** 2:1 **even:odd at every window** — *the cone's thin direction is the window's infrared sector*, fixed in the scale-covariant k-frame, migrating toward DC in physical frequency; radial center $0.31 \to 0.23$ (interior, matching G.4's low-frequency violation anatomy). **(c) The weld**: the window form's bottom cluster at the IR sector *is* the prolate compression — **the thin direction is the CCM prolate sector, rediscovered by an independent instrument from the cone side**, now carrying a measured margin law ($s^* \approx 10^{-1.73\lambda'^2}$, G.4″). The cornering face this delivers: *the open step's geometric content is the uniform interiority of one low-dimensional conditional block — the IR/prolate sector given the rest of the window — with explicit prime-functional entries (the S/D integrals at $k \leq 3$) and the measured budget $s^*(\lambda)$.* Strength conserved (this is the crawl, $\mu_\lambda > 0$ forever, seen geometrically); the identification is now measured rather than argued (P∞-B's allocation verdict, third independent derivation).

12.2 Queue (ii) — the fixed-profile cubic: the equations and the seed measurement

The profile equations. On the seam-only limit profile (§11.2), $d_0 = v_0(1-\rho_w^2) \to v_0$ and the renormalized variable $\hat{R} = R_0\, e^{+p\tau/2}$ is stationary; the renewal's scalar shadow reads $\hat{R}_{n+1} = \hat{R}_n(1 - f_n)e^{+p\Delta\tau/2}$, so stationarity $\iff$ $\langle f_n \rangle \approx p/(2\lambda^2)$ (the measured slide), and the arrival's push decomposes $f_n = \bar{f}(\lambda) +$

$a_1\mathrm{fluc}_n + a_2\mathrm{fluc}_n^2 + a_3\mathrm{fluc}_n^3 + \cdots$ — **the a_k are G.1′'s push-order attribution at the profile, in law-level invariant coordinates where the §8 resolvent disease cannot enter.** G.1(iii)+Littlewood then forbid stationary marginal contact of the law if the windowed a_1, a_3 stay bounded away from zero; the G.2-shaped gaps (identification, windowed persistence) remain and are carried explicitly.

The seed measurement (eleven exact unit-λ^2 pairs, ψ-fluc $\in [-0.43, +0.44]$; registered as a seed, no falsifier). Regression of $\Delta \ln R_0$ on $\{1/\bar{\lambda}^2, \mathrm{fluc}, \mathrm{fluc}^2, \mathrm{fluc}^3\}$: the law term dominates (8 s.e.); the **linear response is unresolved** ($a_1 = -0.14 \pm 0.31$ — the eating does not simply track arrival mass); the first resolved response is **quadratic-positive,** $a_2 = +1.01 \pm 0.38$ **(2.6 s.e.): fluctuation variance of either sign replenishes the seam remainder** (the $3.5 \to 3.64$ inversion at the largest $|\mathrm{fluc}|$ is its visible extreme); the cubic $a_3 = +1.9 \pm 2.1$ is unresolved at $n = 11$, as pre-registered. The chiral channel shows the **opposite variance response** ($a_2 = -7.6 \pm 2.0$ on $\Delta \ln d_-$): *variance feeds the seam and drains the chiral edge* — a two-channel asymmetry registered for the next densification. The 18-window ladder holds $p = 3.039$, $\mathrm{rms}(\log) = 0.081$.

12.3 Queue (iii) — the edge class: the softening is real, the class is open

The $\lambda = 3.6 \to 3.7363$ pair (dps 120, atom-bearing: $n = 13$, fluc $= +0.44$) gives $c_- = 1.79$, $c_+ = 1.76$ — together with round 4's composite step (2.05/1.98), **the phase confound is excluded: the massive-cell constants genuinely soften** beyond $\bar{\lambda}^2 \approx 8$ (certified 2.6–3.3 at $\bar{\lambda}^2 \le 7.8$). Across all twelve pairs, c_- wiggles in 2.6–3.2 then falls to 1.8–2.05, while $p_- = 2\bar{\lambda}^2 c_-$ *rises* $21 \to 52$: **neither pure-exponential nor pure-power class fits the full range** — the edge law's class is open, registered for $\lambda \ge 4$ (dps $\gtrsim 160$). Meanwhile the §11.2 irrelevant flow **confirms dead-on at the new windows**: oddshare$(3.6) = 0.1274$ vs the flow's prediction 0.128; $\rho = -0.434, -0.429$ on the law; the gearing identity at 1.1×10^{-16}; R_{seam} back on the power law after the 3.64 inversion ($f_0 = +0.083$).

12.4 Status: the open step after the queue

Three faces of one ledger, all measured tonight: *geometric* — the IR/prolate conditional block's interiority with budget $s^*(\lambda)$ (12.1); *dynamical* — the seam profile's stationarity with the push-order coefficients seeded and the variance-asymmetry found (12.2); *spectral-law* — p robust at 3.0–3.2 over eighteen windows with the edge class the remaining unknown (12.3). The conserved core is conserved — every face is the crawl in its own coordinates — and the support now consists of: one conditional block, one exponent, one sign ($d_- \ge 0$, certified readable at polynomial cost via G.5.1), with all margins and rates measured. Next-session targets, in leverage order: the **IR-block functional ladder** (the four $k \le 3$ conditional margins as explicit prime sums — the analytic object the thin-direction identification prescribes; its $\lambda \to \infty$ uniformity = the crawl's completion, now in four explicit coordinates); the **densified profile-cubic regression** (more unit pairs; the windowed persistence of a_2's asymmetry and a resolved a_3); the $\lambda \ge 4$ **class run**.

Anchors (§12). §§8–11 (the instruments this section's corrections inherit); CCM 2511.22755 + P∞-B Cor B.3 (the prolate identification's prior faces); F64-5-crossing corollary (banked, 12.1); G.1′/Littlewood (the profile-cubic frame); G.4″ (the budget); `probe_pinfG_thin_direction.py` (v1, corpse) and `_v2.py`; `probe_pinfG_profile_cubic_seed.py`; `probe_pinfG_lambda36_pair.py`; logs and npz archives adjacent.

13. The IR-block functional ladder: the capture theorem-grade weld, the four classical coordinates, and the flat-pace decode of the global ladder

Certificate: `probe_pinfG_ir_ladder.py` + log + `pinfG_ir_ladder.npz` (seven windows $\lambda = 1.8$–3.5, dps 65/110; IR0 GREEN — PSD everywhere, parity cross-blocks of the Gram $\le 5 \times 10^{-43}$; the registered IR1 rule). This is §12.4's leverage-one instrument, executed.

13.1 The object, and the two free facts

For $V_K = \mathrm{span}\{c_0, c_1..c_K, s_1..s_K\}$ (the window's orthonormal Fourier IR vectors), the **conditional form of the IR block given the rest of the window** is $S_{\mathrm{IR}} = \left((K^{-1})_{VV}\right)^{-1}$, computed by one LU and $2K{+}1$ solves; its parity-blocked eigenvalues are the **IR margins**. Two facts are free: *(i)* the kernel rows feeding V_3 are **explicit prime sums** — $P_D[k] = \sum_n \Lambda(n) n^{-1/2}\, 2(1 - \frac{\log n}{L}) \cos(\omega_k \log n)$ and $P_S[k] = \sum_n \Lambda(n) n^{-1/2} \sin(\omega_k \log n)$ at the four slowest frequencies $\omega_k = 2\pi k/L$ — Fejér-weighted, $\sqrt{}$-normalized Chebyshev data, PNT-classical at $k = 0$; *(ii)* interlacing, $\lambda_{\min}(K) \le \lambda_{\min}(S_{\mathrm{IR}})$, so the registered capture rule tests whether the global bottom *lives* in the IR block.

13.2 IR1: the capture weld (the §12.1 identification, theorem-grade measured)

capture ratio $\lambda_{\min}(S_{\mathrm{IR}}^{(K)})/\lambda_{\min}(K)$: $K{=}3 : \le 1.005$ at all seven windows; $K{=}2 : \le 1.053$; $K{=}1 :\ 1.01 \to 1.33$.

The global bottom of the window Weil form *is* the IR-block conditional bottom — to three digits at $K = 3$, and already at $K = 1$ (three dimensions: c_0, c_1, s_1) to within 33% at the largest window. The thin direction's bandwidth is two-to-three IR modes, with a slow λ-drift ($K{=}1$ ratio rising) logged. Consequently the crawl has its promised third face, now certified: **RH's open step = the four-coordinate ladder's uniform interiority** — the parity-blocked conditional margins of one explicit 7×7 object staying positive as $\lambda \to \infty$.

13.3 The conspiracy at coordinate level, and the interleaved ladders

The raw IR coordinates are $O(1)$: $D[0..3] \in [0.037, 1.33]$ across the sweep — while their conditional margins run $4.7 \times 10^{-9} \to 1.2 \times 10^{-52}$: **thirty to fifty decades of cancellation inside one displayable 7×7 Schur complement** — Prop G.7's conspiracy, now localized to four classical coordinates. The margins form two interleaved geometric towers (even 4, odd 3; successive rung ratios $\sim 10^5$–10^6 at $\lambda = 2.69$) — the D1 parity towers seen from the IR side.

13.4 The flat-pace decode (a correction to the ladder's received shape)

Fitting the margins against λ^2 (seven windows):

$$\log_{10} \lambda_{\min} \;=\; -4.85\,\lambda^2 + 7.27 \quad (\text{rms } 0.215),$$

with all four rung families sharing the pace within ± 0.4 $(-4.85, -4.73, -4.60, -4.44)$, rung spacing and parity offset growing slowly ($+0.26\lambda^2$, $+0.12\lambda^2$ in $\log_{10}$). The law reproduces the certified anchors dead-on (ε predictions $-12.1/-16.2/-19.5/-23.1$ vs certified $-12.0/-15.7/-19.1/-22.8$). **The global ladder is a pure exponential in λ^2** — §19.2's $d(s) \sim \exp(-c\, e^{2s})$ with its constant now measured, $c_{\mathrm{ladder}} = 4.85 \ln 10 \approx 11.2$ — **and the "accelerating Fuchs" reading is retired**: it traced to the unmeasured dps-35-era estimate $\epsilon(3.0) \approx 2 \times 10^{-45}$ (true value by this law: $\approx 10^{-36.4}$); the $\lambda = 3.5$ point here is a dps-110 measurement. Consilience: the arrival-direction margin law (G.4″, $-1.73\lambda^2$) runs at 0.36 of the ladder pace — the earlier "0.37 of Fuchs" reading, now between two clean constants.

13.5 The corpses (what the law is *not*), and the analytic target

Two simple models were run against the data before this section and **failed, retained**: *single-gap Slepian leakage* (margins $\sim \lambda^{-2\gamma_1} = \lambda^{-28.3}$ — off by twelve decades at $\lambda = 2.5$ and the wrong shape) and the *fixed-band zero-lattice product* ($\ln \varepsilon$ linear in L — again a λ-power). **The ladder is arrival-paced, not bandwidth-paced**: with matched bandwidth the resolved band is fixed, yet the margins deepen one rung per unit λ^2 — F.7's prime-arrival clock, and (G.7) the pace constant is fluctuation-created. The analytic target this names: *derive* $c_{\mathrm{ladder}} \approx 11.2$ *from the arrival-clocked renewal* — the IR-restricted flow is a 7×7 driven Riccati map with explicit prime driving (each arrival enters the IR rows ramped and phase-coherent,

maturing through the crossover), and the queued **Markov test** — does $S_{\rm IR}(\lambda') \approx F(S_{\rm IR}(\lambda), \text{arrival})$ with the bulk's leak slaved? — would compress the entire crawl to an explicit seven-dimensional flow.

13.6 The locality answer: the conspiracy is full-band (the reduction's second horn)

Certificate: `probe_pinfG_ir_locality.py` + log + npz (L1 GREEN: the truncated conditioning reproduces the T1 exact margins at J = full to $\leq 10^{-16}$). The registered dichotomy resolves **against** the finite-dimensional reduction: $J^*(\lambda)$ — the number of UV frequency shells the IR margins need — **grows to the full band**: $J^* = 4,\ 16,\ 31(=\text{full})$ at $\lambda = 1.8,\ 2.236,\ 2.693$. The convergence profile is near-linear in decades (≈ 1.05 decades of cancellation per shell at $\lambda = 2.69$, 18 decades from raw to exact; the per-shell rate itself grows with λ: $0.3 \to 0.9 \to 1.05$): **every frequency shell participates in the conspiracy** — the $7{+}2J$-dim autonomous flow of §13.5 is a corpse, cleanly measured. The complementarity this leaves is sharp and clarifying: **the chain is local in position (F.21) and global in frequency (this section)** — the renewal's spatial locality and the conspiracy's spectral globality are Fourier-dual faces of the same architecture, and the F-series' position-space renewal coordinates are retro-validated as the only autonomizable frame.

Anchors (§13). §12.1 (the identification this welds); CCM 2511.22755 (the prolate object; $c_{\rm ladder}$ is the measured face of μ_λ's pace); F.7/§19.2 (the arrival clock; the cascade form); G.4″/G.7 (the margin-pace ratio; the fluctuation-created constant); P∞-D §6 (H1′ — the comb machinery the corpses rule out as sufficient); F.21 (the position-local face of the duality); `probe_pinfG_ir_ladder.py`, `probe_pinfG_ir_locality.py`, run logs, npz archives.

14. The densification's honest wash-out, the $\lambda = 4$ pair, and the registered verdict: the power law dies as the asymptote — the renewal decelerates toward a fixed point

Certificates: `probe_pinfG_profile_cubic_dense.py` + log + npz (seven new windows to $\lambda' = 3.868$, all PSD; nineteen exact pairs); `probe_pinfG_lambda4_pair.py` + log + npz (dps 175, dims 97/99 — PSD, d_0-consistency 10^{-111}); `PREREG_lambda4_floor.txt` (the discrimination, **registered before the $\lambda = 4$ numbers existed**); the $\lambda = 4.5/5$ follow-up registered and launched (`probe_pinfG_floor_discriminator.py`).

14.1 The densified regression: the variance signal washes out (GPD self-correction of §12.2)

At $n = 19$ pairs the seed's resolved coefficient collapses: $a_2(\text{seam}) = +0.50 \pm 0.44$ (1.2 s.e., down from 2.6 s.e. at $n = 11$), sign-fragile under the persistence split; the chiral a_2 likewise (-0.50 ± 1.84). **The §12.2 "variance feeds the seam, drains the chiral edge" reading is demoted to unresolved — the seed signal was sample-fragile** (registered PD2 returned "between"; corpse retained). What the densification *does* deliver: (i) the chiral eating's continued collapse ($\Delta \ln d_-$ per arrival: -2.9-grade at $\bar\lambda^2 \leq 8$ $\to -1.07, -1.28$ at $\bar\lambda^2 \approx 14$); (ii) the 27-window ladder ($p = 2.974$, rms 0.082) with the fit-window drift that motivated the registration below: $p(\lambda \geq 1.5) = 2.97$, $p(\geq 2.0) = 2.87$, $p(\geq 2.5) = 2.70$, $p(\geq 3.0) = 2.14$ — **the local exponent falls as the window moves up.**

14.2 The $\lambda = 4$ pair, and the registered read

The falling local exponent admitted two futures, registered with locked thresholds *before* the dps-175 run returned (`PREREG_lambda4_floor.txt`): **(H-pow)** $R_{\rm seam}(4.0) = 0.0229 \pm 8\%$; **(H-floor)** deceleration toward a positive floor, $R(4.0) \geq 0.028$ with pair slope $\ll 2$. The result:

$$R_{\rm seam}(4.0) = 0.02883, \qquad R_{\rm seam}(4.1231) = 0.02884$$

— **flat to four digits across the unit step** ($f_0 = -0.029$: the remainder *rose* through an atom-bearing step, $n = 17$); H-pow missed by +26%, far outside the $\pm 8\%$ wiggle. **The registered read fires for the deceleration class.** And the renewal's coordinates flatten *in concert*: oddshare 0.1123/0.1117 (above the

§11.2 flow's ~ 0.105–0.110 — the flow itself is flattening), $\rho = -0.4146/-0.4141$, $d_-/d_+ \to 1.56/1.55$, and the edge constants collapse to $c_- = 0.90$, $c_+ = 0.89$ (sequence $2.9 \to 2.0 \to 1.8 \to 0.9$; $p_- = 2\bar{\lambda}^2 c_-$ now *falling*, $52 \to 30$ — the edge towers are decelerating too). The gearing identity continues at 5×10^{-17}.

14.3 What died, what opened, and the discrimination in flight

Died: the RG power law as the asymptote — the third rung of the decode chain ($c = 0.342$ was a snapshot of $p/(2\lambda^2)$; $p = 3.2$ was the mid-range law; now $p_{\text{local}} \to\sim 0$ at the top). The §21 determinism argument is **not** contradicted: it forces the *global* ledger $\Sigma\sigma \to 1$, never the *seam cell's* remainder — a seam floor was never structurally excluded, and §23.6's "which clock" question gets its third answer: not per-arrival, not per-e-fold — **the renewal appears to be converging**, with measured candidate fixed-point coordinates $R^* \approx 0.0288$, oddshare$^* \approx 0.112$, $\rho^* \approx -0.414$, $d_-/d_+^* \approx 1.55$: F.25's stationary profile becoming visible in raw coordinates. **Open, with the instrument running:** floor versus log-class decay ($R \sim C/\ln\lambda$ — which is the Śliwiński/CCM crawl's own shape surfacing in the seam) versus residual power; the $\lambda = 4.5/5$ pairs discriminate (locked rules: $R(5.0) \geq 0.0275$ floor; $[0.024, 0.0275)$ log-class; < 0.024 power) — **affordable only because §13.4's flat-pace ladder law cut the dps requirement from ~ 260 to ~ 150.** If the floor is real: the per-arrival eating is *summable* — (U) in its maximal form — and the analytic target becomes proving summability of the seam's per-arrival eating, for which a candidate mechanism is already visible: *fresh arrivals enter the wrap cell Fejér-ramp-suppressed* (SMK-1's geometry), so the seam's eating per arrival may die at a provable rate — with the criterion-vs-arithmetic status of that mechanism the first question to settle. If log-class: the seam remainder obeys the crawl's own law, and the architecture closes on the CCM bound's shape. Either way the open step's quantitative face has moved: **from "one exponent" to "one limit point — is the fixed point interior?" — which is (U) verbatim, now with its coordinates measured.**

Anchors (§14). §§12.2, 11.2, 23.6/§21–22 of P∞-F (the decode chain; the determinism scope); F.25 (the fixed point this measures); §13.4 (the flat-pace law enabling $\lambda = 5$); SMK-1 (the ramp mechanism candidate); GPD (PD2's wash-out retained; the pre-registration discipline); `probe_pinfG_profile_cubic_dense.py`, `probe_pinfG_lambda4_pair.py`, `PREREG_lambda4_floor.txt`, `probe_pinfG_floor_discriminator.py`, logs, npz archives. *(The discriminator's verdict and the correction of this section's fixed-point reading: §15.)*

15. The staircase decode: the seam remainder reads the zeros one at a time

Certificates: `probe_pinfG_floor_discriminator.py` + log + npz (dps 130/150, dims 105–113, all PSD, consistency $\leq 10^{-54}$); the $\varepsilon(4.0)$ backstop (dps 95: 6.6×10^{-66}); `PREREG_lambda4_floor.txt` with its **pre-registered addendum** (the γ-crossing hypothesis and the log-squared candidate, registered before the discriminator returned); the dense-scan test (`probe_pinfG_zero_staircase.py`) registered and in flight.

15.1 The discriminator's data, and the prediction that hit

$$R_{\text{seam}}: \quad 0.02884\ (\lambda^2{=}17) \ \to\ 0.02132\ (20.25) \ \to\ 0.02020\ (21.25) \ \to\ 0.02118\ (25) \ \to\ 0.01909\ (26).$$

The registered buckets returned "residual power" at the $R(5.0)$ threshold — but the **registered log-squared candidate** (addendum, fitted on $R(3.5), R(4.0)$ only, *before* this run) predicted $R(5.0) = 0.0208$: **hit at 2%.** And the fine structure organizes by the zeros: flat across the γ_1–γ_2 gap interior ($16 \to 17$: the §14.3 "floor"); -30% through $\lambda^2 = 17 \to 20.25$ — the γ_2 **coherence-transition zone** (per-arrival phase step $\gamma_2/\lambda^2 \in [1.04, 1.24]$); $+4.9\%$ — *no decay at all* — across the clean γ_2–γ_3 gap ($21.25 \to 25$, 3.75 units, 16% **above the smooth log-squared envelope, beyond the $\pm 10\%$ wiggle**); -10% straddling $\gamma_3 = 25.01$. The backstop adds: the global ladder's pace itself drops ($5.0 \to 3.3$ per unit λ^2) with onset at $\lambda^2 \approx \gamma_1 = 14.13$.

15.2 The correction of §14.3, and the frame

§14.3's fixed-point reading is corrected: the $\lambda^2 = 16$–17 flatness was a *gap-plateau*, not convergence — past γ_2 the coordinates resumed falling (oddshare $0.0986 \to 0.0889$, $\rho = -0.392 \to -0.373$, $c_\pm = 0.90 \to$

0.73). The frame that survives everything measured: **the seam remainder descends a zero-paced staircase.** Mechanism (the registered γ-crossing hypothesis): zero γ_j feeds the seam's innovation while its oscillation is rough on the arrival clock ($\gamma_j/\lambda^2 \gtrsim 1$); as λ^2 sweeps past γ_j the oscillation becomes per-arrival-coherent — locally predictable — and retires. Below γ_1 *every* zero feeds (the strong-eating, power-law-looking regime: the entire old certified range $\lambda^2 \le 12.25$ sits under the first zero); past each γ_j the feeder set thins; the envelope from the $1/\gamma_j^2$-weighted tail is the **log-squared law** $\ln R \approx -c\,(\ln \lambda^2)^2/4$-grade ($c \approx 0.49$ fitted) — $R \to 0$ but $R > 0$ at every finite λ, RH-shaped. The decode chain is now four rungs deep, each the previous law's local read: $c = 0.342 \to p = 3.2 \to$ the deceleration $\to$ **the staircase with log-squared envelope.**

15.3 The zero-detector statement, the honesty, and the test in flight

What this says — and does not. The seam remainder, a positivity-certified $O(1)$ renewal observable computed entirely prime-side, **resolves individual zeros as steps**: γ_1 as the deceleration onset, γ_2's transition zone, the γ_2–γ_3 plateau, γ_3's drop. This is of course the explicit formula speaking (no new arithmetic is created); the deposit is its realization as a *staircase in a certified innovation ledger* — the program's finest decode of its own central observable. Honesty: six points with ±10–15% arrival wiggle make the single-step assignments suggestive rather than settled (only the gap-plateau exceeds wiggle). *(Resolved, same night: the dense scan ran — nine windows, $\lambda^2 = 19.5$–24.25, dps 130, all PSD.* ***The locked rule fired: rate$_A$/rate$_B$*** $= 2.5 \ge 2$ *(zone A* 0.0835 ± 0.046*, zone B* 0.0330 ± 0.036 *per unit* λ^2*), with the d_ tower showing the same directional contrast (*0.93 *vs* 0.76*). Self-attack appended: the registered rule lacked a separation-in-s.e. clause — the contrast is* $\approx 1\sigma$ *against the wiggle, so the staircase is* ***supported at γ_2, not settled beyond wiggle-doubt;*** *the decisive instrument is the multi-crossing scan of §15.4, where steps accumulate statistics across γ_4–γ_7.)* **The open step in staircase coordinates:** $R > 0$ forever $\iff$ no zero's retirement step reaches the remaining remainder — per-zero, local, finite-bandwidth questions ($\delta_j <$ what remains), with the envelope's persistence the conserved core. If the staircase confirms, the per-zero step depths δ_j become the program's new fundamental sequence — measurable, zero-by-zero, against the gap structure of the critical line.

15.4 The per-zero retirement model: the open step, one zero at a time

The staircase frame admits a head-math model whose structure survives the non-circularity audit. Zero $\rho_j = \beta_j + i\gamma_j$ imprints the arrival stream with amplitude $\sim 2x^{\beta_j - 1/2}/|\rho_j|$ in the kernel's normalization; once per-arrival-coherent ($\lambda^2 > \gamma_j$), the renewal predicts the oscillation up to sampling error $\sim (\gamma_j/\lambda^2)^2$ per arrival. For an **on-line** zero ($\beta_j = \frac{1}{2}$, amplitude bounded), integrating the eating across the transition gives a finite step

$$\delta_j \;\sim\; \frac{c}{\gamma_j}, \qquad \sum_j \delta_j \sim c \sum_{\gamma_j \le \lambda^2} \frac{1}{\gamma_j} \sim \frac{c}{4\pi}\,(\ln \lambda^2)^2$$

— reproducing the log-squared envelope from the per-zero side. Calibrating on γ_2's measured step ($\delta_2 \approx 0.30$): $c \approx 6.3$, and the model predicts the **testable law** $\delta_j \gamma_j \approx$ const $\approx$ 5–7 at every subsequent crossing ($\gamma_4 = 30.42,\ \gamma_5 = 32.94,\ \gamma_6 = 37.59,\ \gamma_7 = 40.92, \ldots$ — a crossing-resolved scan to γ_7 is affordable at dps $\lesssim 160$ under the decelerating ladder). For an **off-line** zero ($\beta_j > \frac{1}{2}$) the amplitude grows like $x^{\beta_j - 1/2}$: the retirement never completes, the per-arrival eating does not decay, $\delta_j \to 1$ — the remainder crosses zero at finite radius: F.21's exit, the §14.1 spike, recovered inside the staircase frame. **The audit therefore passes structurally: the per-zero eating law consumes exactly the zero's real part** — the quantity RH is about — through a local, one-zero, finite-bandwidth functional, and the dichotomy is built in: *on-line zeros retire; off-line zeros eat everything.*

The open step in its newest coordinates: RH $\iff$ every zero's staircase step stays below the remaining remainder $\iff$ each $\delta_j < 1$ with the envelope's persistence — per-zero, local statements, with the locality now in the *zeros* rather than in the window. The conserved core is conserved (each $\delta_j < 1$ for the true ζ is what positivity-to-λ certifies, zero by zero, and no finite set of retirements proves the next); what has

changed is the granularity: the open step has gone from one limit, to one exponent, to one staircase — **a sequence of one-zero problems, each with explicit classical structure.**

The multi-crossing run (same night): an instrument kill, and the envelope's longest test. The γ_4–γ_7 scan (`probe_pinfG_multicrossing.py`, twelve windows $\lambda^2 = 28$–41.5, dps 160, dims to 131 — all PSD) returned M2 = 0/4: the per-crossing δ_j extraction is **swamped by a $\pm 20\%$ alternating sawtooth whose period matches the bandwidth-rounding of the matched family** ($N = \mathrm{round}(17.31L)$ increments every $\Delta\lambda^2 \approx \lambda^2/17.3 \approx 2$ — visible as repeated dims in the log; the rounding-induced bandwidth ripple, negligible against the strong mid-range eating, dominates once the per-unit signal is small). Pooled crossing-vs-gap excess: $+0.10 \pm 0.11$ vs -0.04 ± 0.08 — directionally staircase-consistent, $< 1\sigma$. **Instrument kill retained; the corrected design is registered: fixed-N pairs per crossing** (both windows of each local comparison built at the same N, killing the sawtooth within every Δ). The run's clean positive: **M3 — the log-squared envelope holds across three more zero crossings to $\lambda^2 = 41.5$ at rms 0.126 (mean offset $+5\%$)** — now the architecture's best-tested law, spanning $\lambda^2 = 12.25 \to 41.5$ on a two-parameter fit anchored below $\lambda^2 = 17$.

Anchors (§15). §14 (corrected); PREREG file + addendum (the discipline that caught this); §13.4's flat-pace law and its own deceleration (the backstop); F.7 (the arrival clock the coherence mechanism rides); §14.1/F.14 (the off-line spike = the non-retiring zero); N15/Montgomery (zeros-as-spectrum of the prime stream — the classical face); F64-4 (grid/bandwidth artifacts — the sawtooth joins the family); `probe_pinfG_floor_discriminator.py`, `probe_pinfG_zero_staircase.py`, `probe_pinfG_multicrossing.py`, logs, npz.

Anchors. P∞-F §§19–24 (chain, transfer map, F.24–F.27); P∞-D §7 (the RG reading and the lemma queue); P∞-B Thms B.1–B.2 (order dichotomy — the strike's *raison d'être*); P∞-C Prop C.3 (doubling minors); N5 Thm 6.2 ($\tau_n = (-1)^n$); diez §2.6 ($c_2 < 0$ via $L(\frac{1}{2}, \chi_3) = 0.4808\ldots$, LMFDB); E.22.1 (radical mechanism — why the Cramér base is ladder-free); §18/F.19 (pole-matching — the Cramér density's calibration); Littlewood 1914 ($\psi(x) - x = \Omega_\pm$ — the unconditional two-sidedness of §3); CBS/Tower (the order-2 blindness that makes a consumption strike necessary); Connes–Consani, *Weil positivity and trace formula* (2021), Thm 1 (the positive-operator shape welded in §9); CCM arXiv:2511.22755 (the convergence form); GPD (Paper 35); `probe_pinfG_cubic_drift.py` + log + npz; `probe_pinfG_cramer_indefinite.py` + log + npz; `probe_pinfG_cubic_drift_v2.py` + log + npz (§8); `probe_pinfG_sos_bridge.py` + log (§9); `probe_pinfG_increment_bound.py` + log + npz and `probe_pinfG_margin_law.py` + log + npz (§10).